

Homework 2 File Submission

- A. Prove that if x is rational and $x \neq 0$, then $\frac{1}{x}$ is rational.

Hint: Use the definition of a rational. If x is rational, then it can be written as $x = \frac{p}{q}$ where p and q are non-zero integers (written $p, q \in \mathbb{Z}^*$)

- B. Prove using the notion of without loss of generality (WLOG) that $5x + 5y$ is an odd integer when x and y are integers of opposite parity.